Temperature–CVD Project: Exposure File Ready; Defining the HA Analysis

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Laidlaw Scholar — Lab meeting with Professor Bishai, Roro, Hogan, and the team

17 July 2026

1/F Meeting Room, Patrick Manson Building — ∼8–10 minutes

Purpose of this update

Temperature–CVD Project: Exposure File Ready; Defining the HA Analysis

Meeting objectives:

1. Confirm the exposure file and Table 1 environmental panel.
2. Agree on the analysis unit and denominator.
3. Confirm the HA schema, access, and governance steps.

I completed the monthly HKO exposure file and Table 1 Panel A. Roro has confirmed several HA extract features. My proposed default is a monthly population-rate design unless the extract contains a defined at-risk cohort. I would appreciate confirmation on the small set of decisions that determine the first validated HA descriptive report.

What is now confirmed

Exposure (completed)

- HKO Headquarters, Jan 2013–Dec 2023
- $N = 132$ months
- Monthly mean / Tmin / Tmax and extreme-day counts
- Validated against HKO *Year's Weather* (33/33)

HA extract (Roro has confirmed)

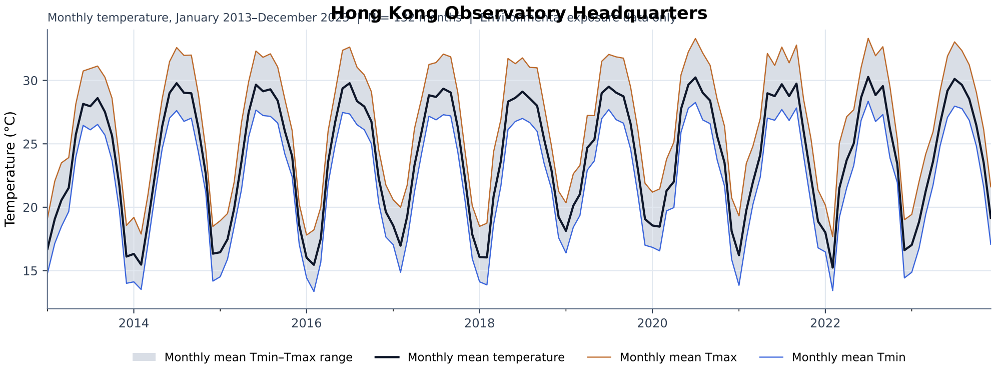
- ICD-9
- Patient-level admission information
- Ages 65–69 and 70–74 available separately
- No ED-to-inpatient episodes included

Still pending before real admission analysis

- Wet-ink signatures / HPC access (Bob + Hogan; Roro to onboard)
- No real admission analysis yet — exposure and design only today

Grateful to Roro for the 12 July clarifications. Remaining questions are definitional (DAE, denominator, meds/BMI timing), not a gap in the reply.

Temperature visualization — HKO Headquarters



Environmental exposure data only. Series ends December 2023 ($N = 132$ months).

Table 1 structure — two panels, two denominators

Panel A. Monthly environmental exposures

$N = 132$ months HKO Headquarters

Variable	N	Mean	SD
Mean temperature (°C)	132	24.02	4.70
Mean Tmin (°C)	132	22.10	4.69
Mean Tmax (°C)	132	26.66	4.82
Hot nights (days/month)	132	3.40	5.40
Cold days (days/month)	132	1.10	2.55
Very hot days (days/month)	132	3.19	5.18

Panel B. Patient / admission characteristics

Pending HA access — denominator TBD

Requested fields (descriptive Table 1 as directed): number of patients; qualifying episodes; age; sex; diuretics; beta blockers; metformin; SGLT2 inhibitors; BMI and BMI missingness.

Panel A has 132 month-level observations. Panel B will use a patient-, admission-, or patient-month-level denominator. I will not place both under one undifferentiated N. Medication/BMI model role depends on whether a true at-risk cohort exists and whether measurements precede the event.

Analysis design to confirm

Proposed default: monthly population-rate design

Patient-level HA records

Define qualifying episodes

Aggregate by month × age × sex × diagnosis

Merge HKO exposures + C&SD person-time

Negative Binomial monthly admission-rate model

Alternative, if available

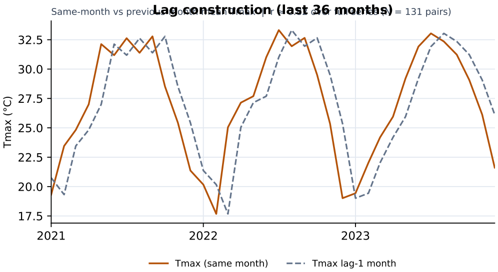
If a defined cohort with non-event follow-up exists

Patient-month risk model

Admissions-only extracts support **monthly admission counts / rates**, not "patient-level risk," until the denominator is defined.

My proposed default: retain the monthly population-rate design unless the HA extract contains a defined at-risk cohort. I would appreciate confirmation.

Lag variables and proposed model ladder



- Contemporaneous and lag-1 variables can both be added to the merge.
- Current and lag-1 Tmax are strongly correlated ($r \approx 0.82$; 131 pairs).
- Same-month exposure proposed as primary.
- Lag-1 proposed as a separate sensitivity model.
- Current/previous-month average may be secondary.
- Do not put mean, Tmin, Tmax, and all lags together in one initial model.

December 2012 daily HKO data exist in the raw extracts, but the processed monthly file currently starts January 2013, so January 2013 lag-1 is missing unless we extend the weather build. I recommend extending the weather input to December 2012.

I propose this model ladder and would welcome confirmation.

Confirmations needed and next deliverable

1. **DAE and row/episode definition.** What “DAE records included” means; whether one row = one admission episode.
2. **Principal diagnosis, admission route, transfers, and recurrence.** Principal-only vs any position; elective vs emergency; transfer/recurrence rules.
3. **Admissions-only dataset versus defined at-risk cohort.** This determines whether we stay with monthly population rates or move to a patient-month risk model.
4. **Medication/BMI timing and IRB amendment.** Are fields on the approved extract? Pre-event timing? IRB update is a PI decision.

Immediate actions

- **Bob:** wet-ink; extend weather to Dec 2012; prepare HA descriptive scripts; draft Table 1 Panel B shell.
- **Roro:** clarify DAE / episode / denominator / meds-BMI fields; onboard Bob after signatures.
- **Hogan:** wet-ink; advise on station choice / any pollution extension timing.
- **Professor Bishai:** confirm design default, model ladder, covariate role, and IRB path.

Next deliverable: validated HA descriptive counts, missingness, episode definitions, age-specific rates, and completed Table 1. No causal claims.

Appendix

Technical detail for discussion if needed

A1. Provisional ICD-9 definitions

Coding system **confirmed ICD-9** (Roro, 12 July). Code lists remain provisional until principal-diagnosis and local HA rules are confirmed.

Group	Codes	Notes
AMI	410.x	Exclude old MI 412
Hemorrhagic stroke	430, 431; provisionally 432.x	Sensitivity: 430–431 only
Ischemic stroke	433.x1, 434.x1, 436	Sensitivity: with/without 436
Exclude	412; 435.x; 438.x	Old MI; TIA; stroke sequelae

Source: `memos/data_request_roro.md`. ICD-10 backup retained only if needed historically.

A2. Full HKO validation table (annual extremes)

Station: HKO Headquarters. Result: **33/33 MATCH** vs HKO *The Year's Weather*.

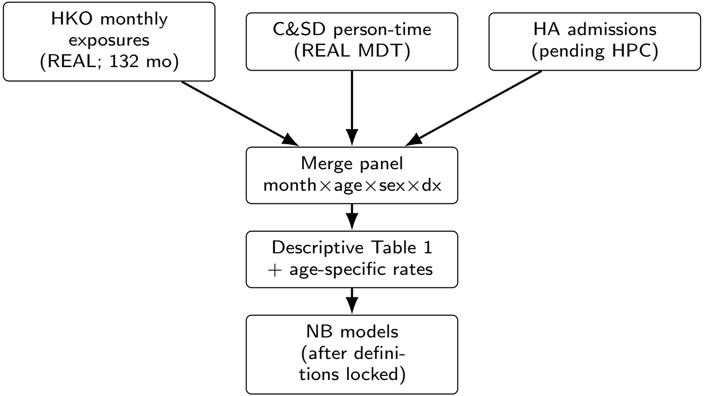
Year	Hot nights	Very hot days	Cold days
2013	10	17	14
2014	34	33	21
2015	37	28	7
2016	36	38	21
2017	41	29	9
2018	26	36	21
2019	46	33	1
2020	50	47	11
2021	61	54	13
2022	52	52	13
2023	56	54	14

Source: `outputs/tables/hko_annual_extremes_validation.csv`.

A3. Detailed data questions for Roro

1. What does “DAE records included” mean in this ICD-9 extract? (I do not invent an expansion.)
2. Is each row one admission episode, with drugs/BMI attached if available?
3. Can principal diagnosis be distinguished from secondary diagnoses?
4. Are elective inpatient admissions included, given no ED-to-inpatient episodes?
5. How are inter-hospital transfers and same-patient recurrence handled?
6. Are diuretics, beta blockers, metformin, SGLT2is, and BMI already on the approved extract?
7. What medication and BMI measurement dates are available relative to admission?
8. Does the extract contain non-event follow-up / a defined at-risk cohort, or admissions only?
9. What small-cell suppression rules apply to descriptive outputs from HPC?

A4. Pipeline diagram (ready for real HA merge)



Scripts: `02_build_weather_monthly.R`, `05_build_population_denominators.R`, `08b_merge_real_ha_panel.R`, `09–11`. Synthetic track exists only for workflow testing — not findings.

A5. Full Table 1 shell

Panel A — filled ($N = 132$ months; HKO Headquarters)

Mean temp 24.02 (4.70); Tmin 22.10 (4.69); Tmax 26.66 (4.82); hot nights 3.40 (5.40); cold days 1.10 (2.55); very hot days 3.19 (5.18).

Panel B — pending HA access

Field	Status / note
N patients / episodes	Pending; denominator depends on design fork
Age, sex	Requested; strata include 65–69 and 70–74
Diuretics; beta blockers	Requested descriptively; model role TBD
Metformin; SGLT2 inhibitors	Requested descriptively; model role TBD
BMI (+ missingness)	Requested descriptively; pre-event timing TBD

I will include these covariates descriptively as requested. Their use as model confounders depends on cohort structure and measurement timing.

A6. Possible pollution and district extensions

- **Pollution:** EPD import scripts ready; current file is a placeholder. Near-term priority remains temperature → AMI/stroke after HA access. Pollution can enter staged models after the first validated descriptive report.
- **District / multi-station weather:** HKO Headquarters is the primary station (matches official extreme-day definitions). Multi-station or district extensions are optional sensitivities (Hogan), not required for the first HA report.
- **2024 weather extension:** Optional; raw 2024 extract exists. Do not delay the 2013–2023 HA core.